**Please check the examination details below before entering your candidate information**

| Candidate surname | Other names |
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| Centre Number | Candidate Number |
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# Pearson Edexcel GCSE (9–1)

## Mock Examination (TEACHER MARK SCHEME & EXEMPLARS)

Morning (Time: 1 hour 20 minutes)

**Paper reference: 1HI0/11**

## History

**PAPER 1: Thematic study and historic environment**

**Option 11: Medicine in Britain, c1250–present and The British sector of the Western Front, 1914–18: injuries, treatment and the trenches**

**You must have:**
Sources Booklet (enclosed)

Total Marks

### Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- There are two sections in this question paper. Answer all parts of Questions 1 and 2 from Section A. From Section B, answer Questions 3 and 4 and then **EITHER** Question 5 **OR** Question 6.
- Answer the questions in the spaces provided – *there may be more space than you need.*

## Information

- The total mark for this paper is 52.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
- The marks available for spelling, punctuation, grammar and use of specialist terminology are clearly indicated.

## Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

## SECTION A: THE BRITISH SECTOR OF THE WESTERN FRONT, 1914–18: INJURIES, TREATMENT AND THE TRENCHES

**1 (a) Describe one feature of the clinical use and diagnostic limitations of mobile X-ray units near the frontline.**

(2)

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**1 (b) Describe one feature of trench fever, including its symptoms, cause, or prevention.**

(2)

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**2 (a) Study Sources A and B in the Sources Booklet. How useful are Sources A and B for an enquiry into the physical trauma and preventative treatments associated with German poison gas attacks? Explain your answer, using Sources A and B and your knowledge of the historical context.**

(8)

**2 (b) Study Source B. How could you follow up Source B to find out more about the preventative treatments associated with German poison gas attacks? In your answer, you must give the question you would ask and the type of source you could use. Complete the table below.**

(4)

| |
|---|
| Detail in Source B that I would follow up: |
| Question I would ask: |
| What type of source I could use: |
| How this might help answer my question: |

**(Total for Question 2 = 12 marks)**

## SECTION B: MEDICINE IN BRITAIN, C1250–PRESENT

**3 Explain one way in which ideas about the cause of illness in the Medieval period (c1250–c1500) were similar to ideas about the cause of illness in the Renaissance period (c1500–c1700).**

(4)

**4** **Explain why there was rapid progress in the development and mass production of penicillin in the years c1938–c1945.**

> You **may** use the following in your answer:
>
> - Howard Florey and Ernst Chain
> - US Government funding and the Second World War
>
> You **must** also use information of your own.

(12)

**MODEL ANSWER (GRADE 9)**

The primary scientific driver behind the rapid progress of penicillin was the systematic biochemical work of Howard Florey and Ernst Chain at Oxford University, who succeeded in turning Alexander Fleming's 1928 accidental laboratory observation into a purified, stable, and highly effective drug. While Fleming had discovered that the Penicillium notatum mould killed staphylococci bacteria, he lacked the chemical expertise to isolate the active substance and abandoned it as a minor local antiseptic. In 1938, Florey and Chain revived the research, and by 1940, Chain successfully purified the active compound. Consequently, they demonstrated penicillin's curative power on infected mice, followed by successful trials on human patients, such as police officer Albert Alexander in 1941. This clinical proof demonstrated that penicillin was non-toxic and could destroy bacterial pathogens inside the bloodstream, establishing the scientific foundation necessary to justify transition to mass production. This scientific breakthrough was scaled up globally because of unprecedented US Government funding and direct state intervention under the urgent pressures of the Second World War. When Florey travelled to the USA in July 1941 to seek manufacturing aid, British chemical companies were fully occupied with munitions production. Recognizing the immense military value of an infection-killing antibiotic for treating battlefield casualties and preventing fatal gas gangrene, the US government bypassed traditional market constraints and poured millions of dollars in federal grants into pharmaceutical firms. Consequently, this state-funded mobilization allowed American factories to purchase expensive industrial infrastructure, including giant deep-fermentation tanks. This massive financial and logistical intervention led directly to a manufacturing revolution, allowing the supply of penicillin to scale up from virtually zero to over 2.3 million life-saving doses in time to treat Allied forces on D-Day in June 1944. Furthermore, progress was accelerated by international scientific collaboration and advances in agricultural technology, which provided a distinct third factor beyond the initial stimulus points. To maximize production yields, researchers at a government-backed laboratory in Peoria, Illinois, systematically searched for more productive strains of the Penicillium mould, discovering a highly potent, high-yield strain on a cantaloupe melon in 1943. By combining this new strain with the deep-fermentation technology, scientists multiplied the production rate of pure penicillin millions of times over. This meant that bacterial infections, which had previously been the leading cause of death following minor cuts or surgeries, became completely curable on a mass scale, permanently transforming modern global healthcare and establishing penicillin as the world's first mass-produced antibiotic.

**Answer EITHER Question 5 OR Question 6. Spelling, punctuation, grammar and use of specialist terminology will be assessed in this question.**

**EITHER**

**5** **'Edward Jenner's discovery of the smallpox vaccination was the most significant turning point in the prevention of disease in the period c1700–c1900.' How far do you agree? Explain your answer.**

> You **may** use the following in your answer:
>
> - Inoculation
> - The 1875 Public Health Act
>
> You **must** also use information of your own.

(16)

**OR**

**6** **'The discovery of the structure of DNA was the most significant breakthrough in understanding the causes of illness in the period c1900–present.' How far do you agree? Explain your answer.**

> You **may** use the following in your answer:
>
> - James Watson and Francis Crick
> - Lifestyle factors (e.g., smoking and lung cancer)
>
> You **must** also use information of your own.

(16)

(Total for spelling, punctuation, grammar and use of specialist terminology = 4 marks)

**(Total for Question = 20 marks)**

**Indicate which question you are answering by marking a cross in the box [X].**

Chosen question number: Question 5 [ ] Question 6 [ ]

**QUESTION 5 MODEL ANSWER (GRADE 9)**

To a moderate extent, I agree that Edward Jenner's development of the smallpox vaccine in 1796 represents a monumental turning point in the prevention of disease. By utilizing empirical observation to prove that cowpox infection conferred immunity against smallpox, Jenner created the world's first secure, scientific immunisation technique. However, while his discovery was intellectually revolutionary, its practical significance was heavily limited in the short term by public conservatism and medical opposition. Ultimately, the transition to truly effective, nationwide disease prevention was only achieved when the British government abandoned its traditional laissez-faire policy and passed compulsory public health legislation, most notably the 1875 Public Health Act, which systematically tackled urban filth. On the one hand, Jenner’s 1796 smallpox vaccine was a major turning point because it provided a safe, scientific alternative to the dangerous practice of inoculation. Inoculation, which had been introduced to Britain from Turkey in the early eighteenth century, involved deliberately scratching live smallpox pus into a healthy person's skin. While this often conferred mild immunity, it was highly dangerous: it frequently caused full-blown, fatal infections and actively turned inoculated individuals into contagious vectors, sparking new epidemics. By vaccinating young James Phipps with benign cowpox and subsequently proving he was immune to smallpox, Jenner demonstrated that prevention did not require risking the patient’s life. Consequently, Jenner’s work provided a safe, reliable, and reproducible medical tool that eventually enabled the World Health Organization to eradicate smallpox globally, representing a profound milestone in immunology. However, the immediate preventative significance of Jenner's discovery was heavily undermined by deep-seated public conservatism, medical self-interest, and the absence of a scientific framework to explain how vaccines worked. Because Jenner published his findings in 1798 without being able to explain *why* cowpox prevented smallpox—as Germ Theory was not published until 1861—many doctors and citizens regarded vaccination with extreme suspicion. Anti-vaccination campaigns published satirical cartoons of people turning into cows, and the Royal Society refused to publish his research, forcing him to print it privately. Furthermore, inoculation was a highly lucrative business for professional doctors, who actively lobbied against the free adoption of Jenner's vaccine to protect their profits. As a result of this resistance, smallpox remained a major urban killer for decades after Jenner’s breakthrough, demonstrating that a scientific discovery alone is insufficient to prevent disease without administrative force. Furthermore, the primary driver of rapid, nationwide disease prevention during the nineteenth century was not medical discovery, but the British government’s complete abandonment of its laissez-faire administrative policy. Spurred by John Snow's 1854 proof that cholera was water-borne and Bazalgette's massive London sewer construction, Parliament passed the landmark 1875 Public Health Act. Unlike previous optional acts, the 1875 Act made sanitation compulsory. Local councils nationwide were legally forced to provide clean, filtered water, build modern underground drainage, clear rubbish, and employ medical officers.

Consequently, by systematically removing the filthy, contaminated water and organic waste that caused water-borne epidemics, the government dramatically reduced the national mortality rates of diseases like cholera and typhoid. This administrative intervention saved far more lives than Jenner’s vaccine did in the nineteenth century, proving that state power was the true engine of prevention. In conclusion, while Jenner’s vaccine was a brilliant theoretical breakthrough that laid the foundation for modern immunology, it was only a partial turning point c1700–c1900. Without the scientific validation of Louis Pasteur’s Germ Theory (1861) and, crucially, the administrative enforcement of the compulsory 1853 Vaccination Act, Jenner's work remained underutilized. The most significant turning point in disease prevention during this era was the compulsory 1875 Public Health Act. By utilizing state power to enforce urban sanitation, the government actively addressed the environmental causes of epidemics, demonstrating that administrative intervention was far more effective at protecting public health than isolated scientific breakthroughs.

# Sources Booklet

Do not return this Booklet with the question paper.

**Sources for use with Section A.**

**Source A: A clinical extract from an RAMC medical officer logbook at an Advanced Dressing Station (ADS) near Ypres, May 1915.**

> The men were brought in stumbling, clutching their throats and gasping violently for air. Their eyes were severely inflamed and streaming with tears. We immediately began washing out the blinded eyes of the chlorine gas casualties using simple bicarbonate of soda solutions. It provides some relief, but the respiratory damage is profound. Many will not survive the night.

**Source B: A photograph showing Belgian troops wearing early gas masks.**

> Visual Source
>
> The photograph shows a group of Belgian soldiers standing outdoors, wearing early protective flannel gas hoods pulled over their heads to protect against chlorine gas attacks.